So, $(a+b)^2 = a^2 + ba + ab + b^2$

$= a^2 + ab + ab + b^2$ (since, $ba = ab$)

$(a+b)^2 = a^2 + 2ab + b^2$

Hence proved.

Note

If, we substitute $b = a$ in the identity $(x+a)(x+b) = x^2 + x(a+b) + ab$, we get,

$(x+a)(x+a) = x^2 + x(a+a) + a \times a$

$(x+a)^2 = x^2 + x(2a) + a^2$

$(x+a)^2 = x^2 + 2ax + a^2$, which is similar to the identity $(a+b)^2 = a^2 + 2ab + b^2$.

Example 3.2 Simplify the following using the identity $(a+b)^2 = a^2 + 2ab + b^2$.

(i) $(2x+5)^2$ (ii) 21^2

Solution

(i) $(2x+5)^2$

Let the side of the square be $2x+5$ units. Then its area is *side* $\times$ *side*, that is $(2x+5)^2$.

The geometrical representation of the given expression is as shown in *Fig. 3.15*.

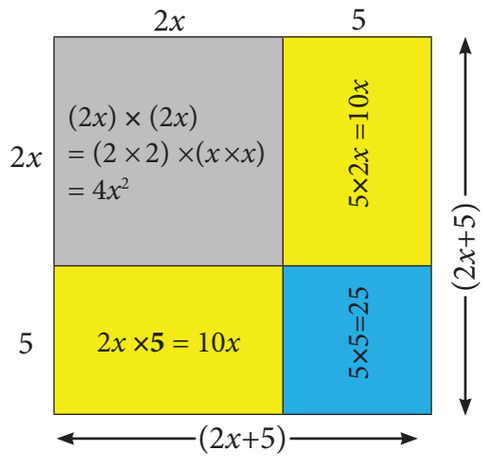


Fig. 3.15

Area of the bigger square = **Area of two squares + Area of two rectangles.**

$(2x+5)^2 = 4x^2 + 25 + 10x + 10x$

$= 4x^2 + (10+10)x + 25$ (Adding like terms)

$= 4x^2 + 20x + 25$.

(ii) 21^2

Let the side of the square be 21. Hence, its area is 21^2.

Consider 21^2 as $(20+1)^2$ which is one of the way to represent it geometrically as shown in *Fig. 3.16*.

Now,

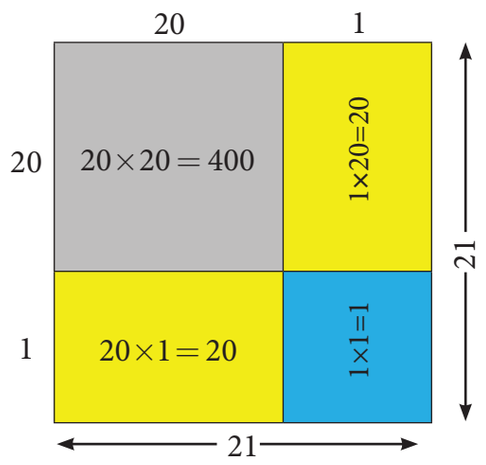


Fig. 3.16

Area of the bigger square = **Area of two squares + Area of two rectangles.**

$21^2 = 400+1+20+20=441$.

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